



VECTOR SPACES

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CLASS 6 : CONTENT



- Column Space
- Row Space

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COLUMN SPACE



The column Space

Definition: Let A be a $m \times n$ matrix. The column space of A is the set of all linear combinations of the columns of A denoted by $C(A)$. Thus,

$$C(A) = \{ b \in \mathbb{R}^m / Ax = b \text{ is solvable} \}$$

Note : $C(A)$ is a subspace of $\mathbb{R}^m$

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- In other words, The space spanned by linear combination of linearly independent columns of matrix A spans the Column Space of Matrix A .
- Column Space is denoted by $C(A)$
- $C(A)$ can lie anywhere in between the zero space and the whole space $\mathbf{R}^m$
- The system of Linear equations $Ax=b$ is solvable iff the vector ' b ' can be expressed a combination of columns of A , then ' b ' is in $C(A)$.

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The column Space

Few examples.... 1. The smallest possible column space comes from the zero matrix $A = 0$. The only combination of the columns is $b = 0$.

2. If A is a 5×5 identity matrix then $C(A)$ is the whole of $\mathbb{R}^5$ the 5 columns of A can combine to produce any 5 dimensional vector b . In fact, any 5×5 nonsingular matrix A will have $\mathbb{R}^5$ as its column space !!

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Let

$$A = \begin{bmatrix} 1 & 0 \\ 5 & 4 \\ 2 & 3 \end{bmatrix}$$

then $C(A)$ is the subspace of $\mathbf{R}^3$ consisting of vectors b that are linear combinations of the vectors $(1, 5, 2)$ and $(0, 4, 3)$. Geometrically the subspace is a 2- d plane.

Let

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 5 & 4 & 9 \\ 2 & 3 & 5 \end{bmatrix}$$

Then $C(B)$ is the subspace of $\mathbf{R}^3$ consisting of vectors b that are linear combinations of the vectors $(1, 5, 2)$, $(0, 4, 3)$ and $(1, 9, 5)$.

COLUMN SPACE



Note : The column spaces of A and B are same though the matrices are different. This is because the new column is a linear combination of the other two columns. Hence, appending a dependent column does not alter the column space of a matrix .



THANK YOU

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